

Repressive Democracies: State Repression, Protest, and Policy Change

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Abstract

Democratic governments decide in advance how harshly police may treat protesters, and preparing for a large crowd is expensive. Much is known about whether repression deters or provokes protest, less about how the cost of managing a crowd affects authorized harm. This paper models an elected executive who sets a policing directive before citizens decide whether to join a protest that succeeds only if enough participate. A harsher directive discourages participation and lets the state prepare for a smaller crowd, but raises legal, electoral, and operational costs. When preparing for each protester becomes costlier, the state authorizes more harm if the resources saved by deterring participants exceed the added cost of harsher tactics. Budget pressure therefore produces harsher repression where policing costs lie in deployment, and restraint where the same inputs implement force. Stronger accountability lowers expected harm. Repression backfires only when motivation rises faster than expected harm.

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1 Introduction.

Mass protests are costly for incumbent elites in democracies for two reasons. A massive protest may force politicians to change policies they prefer to preserve (Bosi, Giugni, and Uba 2016; Klein and Regan 2018). Massive protests also require the state to deploy personnel, equipment, transport, detention capacity, and other operational resources, and these resources must be planned before turnout is known. Elected executives and police commanders set deployments, authorized tactics, and rules of engagement for anticipated protest episodes, within the framework set by legal rules, standing procedures, and political oversight. A harsher directive can deter participation and thus reduce the crowd the state must prepare for. This paper asks whether the cost of managing a crowd affects how severely a democratic state polices protest. Specifically, we ask whether a government authorizes more harm against protesters, or less, when preparing for them becomes more expensive.

Officers and commanders already weigh this tradeoff. Greek riot police interviewed by Vasilaki (2016, p. 20) name numerical shortfall directly: “Operations are designed in a way that makes us violent. We are few against many.” British officers interviewed after the 2011 riots “said it was a lack of resources that resulted in them losing control,” and the Metropolitan Police afterward sought water cannon and authorized baton rounds (O’Sullivan 2017, pp. 35–36). Yet the same budget pressure also produced restraint, since the Metropolitan Police extended protest liaison partly for “the possibility it can present for reducing the amount of police resources that are mobilised at protests” (p. 198). These accounts show that the question arises in practice, but they cannot answer it. The crowd is not fixed in advance, since a harsher directive discourages some citizens from joining. Hence, the state’s choice changes the quantity it is preparing for, and the direction of the

budget effect has to be derived.

Studies disagree about how state repression affects mobilization (Davenport 2007a). Some find that threats of repression deter mobilization by raising the cost of protest (Chenoweth, Perkoski, and Kang 2017; della Porta 1995; C. M. Sullivan 2016). Others find that repression can backfire by radicalizing victims of state violence (Adam-Troian, Çelebi, and Mahfud 2020; C. M. Sullivan and Davenport 2017). Scholars have studied this nexus between protest and repression primarily in autocracies (Davenport 2007a; Curtice and Behlendorf 2021). An empirical strand examines protest in democracies (Carey 2006; Nam 2007; Machado, Scartascini, and Tommasi 2011). It asks mainly why protest occurs more often in some democratic settings than in others (Su 2015; Aytaç, Schiumerini, and Stokes 2017). Competitive elections, protected speech, and organized civil society make protest a viable complement to representative channels, while legal and electoral accountability make the costs of coercion especially visible. Democratic protest policing therefore places deterrence, operational capacity, and political accountability in the same strategic problem.

Neighboring literatures have not reached this question. Research on protest policing explains the choice of tactics mainly through threat perception, security diffusion, and legal change (Noakes and Gillham 2007; Gillham 2011; Selmini and Di Ronco 2023). None of these accounts discusses police budgets. Studies of capacity and fiscal pressure come closer. H. Sullivan (2021) finds that Mexican subnational governments with bureaucratic capacity respond to protest with repression less often. Beck (2023) finds that United States municipalities that rely more on fines and fees experience more police killings, a fiscal determinant of coercive severity outside the protest setting. Bueno de Mesquita and Shadmehr (2023) provide the closest application of global games to repression, and their government pays a single cost of its repression level. That cost does not vary with the size of the crowd, and the model has no separate political cost of authorizing harm. To our knowledge, no study asks whether fiscal constraint on the policing apparatus changes the

severity of authorized tactics.

To study this question, we develop a sequential model in which citizens protest to change a policy. The elected executive first sets the expected repression cost borne by a participant. This measure combines the reach and intensity of coercion. A continuum of citizens then decides whether to protest. The protest changes policy only if participation reaches an uncertain critical threshold, and each citizen observes a noisy private signal of that threshold. Participants receive an expressive reward only if the protest succeeds, and they pay the expected repression cost. The state values preservation of the disputed status quo. It plans operational resources for the expected volume of protest and pays a convex political cost from expected participant harm. We take the participation stage from the psychological rewards model of Bueno de Mesquita and Shadmehr (2023) and give the state a payoff that separates these two costs.

The central result is a condition on how the police organization produces force. When preparing for each additional protester becomes more expensive, the state authorizes more expected harm if the resources saved by deterring participants exceed the added cost of implementing harsher tactics. The logic has two steps. Expected harm discourages some citizens from joining, and a smaller expected crowd lets the government plan fewer officers, vehicles, detention spaces, and supplies. Thus, budget pressure produces harsher repression where policing costs lie mainly in deployment. Yet the same budget pressure can instead produce restraint if stronger tactics require specialized officers, equipment, documentation, medical support, or detention capacity. What matters is the cost of managing one more protester. The total price of maintaining a police force is a different quantity. Additionally, stronger accountability lowers expected harm. An extension shows that repression backfires only when protesters' motivation rises faster than the harm they expect, and that a state anticipating a stronger emotional response authorizes less harm when its costs lie in deployment.

The model also tells empirical researchers what to measure. First, expected participant

harm combines reach, the probability that a participant is targeted, with the harm conditional on targeting, and event records can code both. Second, researchers can distinguish standing capacity from resources added in anticipation of protest, using budget documents, deployment plans, procurement records, and reports written after each episode. The same records can show whether a cost change falls mainly on deployment or on implementing force, which determines the sign of the budget effect. Third, survey and experimental designs can measure the reward citizens expect from a successful protest and how it responds to expected harm.

The rest of the paper reviews the literatures on protest, repression, and coordination, presents the participation stage, develops the state's problem and its comparative statics, and discusses the paper's theoretical and empirical contribution.

2 Literature review.

2.1 State repression, political attitudes, and social movements.

Research on state repression examines why governments coerce citizens and how coercion affects political behavior, especially protest. Political repression is the use of institutional or physical force, most commonly violent, to subdue or coerce individuals and groups that challenge political actors or the status quo (deMeritt 2016). Much of this research focuses on authoritarian regimes or flawed democracies, where rulers face weak accountability and citizens lack protected speech or competitive elections.

Comparative research has persistently found that democratic countries exhibit lower levels of state repression (Davenport 2007a, 2007b; Davenport and Armstrong 2004). Yet violent coercion by state agents also occurs in developed democracies (Arias and Goldstein 2010; Belur 2010; Worden 1995). Citizen backlash is not confined to force used against protesters. George Floyd was killed during an arrest, not while participating in a demonstration. The ensuing protests reduced favorability toward the police among

low-prejudice and politically liberal Americans (Reny and Newman 2021). Police violence outside a protest can therefore mobilize citizens and raise the political costs that surround later police action. Victims of police repression may also experience anxiety, depression, anger, and fear (Bor et al. 2018; Çelebi, Adam-Troian, and Mahfud 2022; Cooper et al. 2004; J. E. DeVolder et al. 2017; J. DeVolder, Fedina, and Link 2020).

Yet not all citizens perceive the use of force by repressive bureaus the same way. In the United States, for example, studies consistently find that conservatives support the police at higher rates than liberals (Ekins 2016; Fingerhut 2017; Hindelang 1974; Zamble and Annesley 1987). Voters may perceive and evaluate state repression differently, to the extent that they have directional motivations and are affective partisans. Attitudes about the police are also racially divided (Ekins 2016; Fingerhut 2017).

Davenport (2007a) calls the conflicting evidence about how repression affects social movements the “punishment puzzle.” Some studies find that repression discourages mass and peaceful protest but encourages radicalization (della Porta 1995). Others find that it increases both protest escalation and mobilization (Adam-Troian, Çelebi, and Mahfud 2020; C. M. Sullivan and Davenport 2017).

2.2 Empirical regularities and theoretical foundations.

The literature has identified several determinants that systematically explain state repression, among them regime type, economic development, population size, intrastate civil conflicts, and the degree of grievances and dissent (Davenport 2007b; Nordås and Davenport 2013; C. M. Sullivan 2016; C. M. Sullivan and Davenport 2017). For example, Nordås and Davenport (2013) argue that the demographic structure of the population is a key determinant of state repression. States respond with more repression where youth bulges are larger, meaning cohorts aged 15 to 24 relative to the overall adult population. States that anticipate more dissent raise repression preemptively, to deter it.

Chenoweth, Perkosi, and Kang (2017) summarize several consistent findings in the

literature. First, dissent precedes state repression in most settings, a regularity Davenport (2007a, p. 7) calls the “law of coercive responsiveness.” Second, democratic countries tend to exhibit lower levels of repression. The authors also discuss the “punishment puzzle” mentioned above. They stress that repression and dissent, whether protest or other tactics, are two complex interlinked processes.

Formal models explain the relationship between state repression, protests, and regime or policy change in several ways. No definite consensus has emerged about the best modeling approach. Ritter (2014) notes that models choose different objects of investigation, and that each one, taken alone, misses important strategic features of the relation between repression and dissent. For example, some studies model the determinants of regime change or dissent in general as a process of an ongoing or expected conflict. In one approach, citizens protest to alter the status quo, and their success depends on the state’s concessions. A revolutionary vanguard’s level of violence may signal the level of dissent in society, which provokes citizens to cooperate (Bueno de Mesquita 2010). Other formal models turn to uncertainty and asymmetric information about government types (Pierskalla 2010). A “strong” state may repress citizens at high cost to signal its resolve. Citizens protest over repeated play and learn what type of government they face. Closest to the present paper in timing, De Jaegher and Hoyer (2019) model *preemptive* repression. A government sets repressive capacity before protest occurs, and dissidents respond by investing in countermeasures. De Jaegher and Hoyer derive both sides of the punishment puzzle. Repression deters when the benefit of protest is large relative to the cost of countermeasures, and backfires when it is small. Their complete-information coordination game has two dissidents, each of whom can determine the outcome. The relative size of the equilibria’s basins of attraction resolves the resulting multiplicity.

Thus, as Ritter (2014) notes, some models study the conditions that ignite dissent, and others study how the state chooses the severity of repression. But studies miss an important part of the puzzle when they treat these two processes separately. She therefore

proposes a model that considers the conditions sparking dissent together with the state's choice of severity. She models the strategic setting as a bargaining problem over policy provision and its effect on incumbent survival. Ritter finds that a higher likelihood of leader survival decreases the level of dissent and repression. Yet the same increase raises the severity of both.

2.3 Global games of protest and regime change.

This paper borrows its analytic setup from the global games literature on coordination under incomplete information (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). The canonical regime change game has a continuum of citizens who simultaneously decide whether to attack a regime. The regime collapses if and only if the mass of attackers exceeds its strength. That strength is an uncertain fundamental, and citizens receive noisy private signals about it. Under complete information about the fundamental, the game exhibits multiple equilibria in an intermediate range of regime strength. The incomplete information selects a unique equilibrium in threshold strategies.

Bueno de Mesquita and Shadmehr (2023) provide the closest application of this framework to repression. Their game with psychological rewards is “a standard global game of regime change” (p. 738). A continuum of citizens receive noisy private signals $x_i = \theta + \sigma \varepsilon_i$ about the regime's strength θ , and the rebellion succeeds if and only if the fraction of rebels exceeds θ . Their psychological-rewards setting yields the threshold $1 - c$, where c is the participation cost, and their government chooses costly repressive capacity before citizens move. The present paper takes that participation stage as its benchmark. Its contribution begins with the government's payoff. The state values a disputed policy status quo rather than regime survival, and it separately accounts for the operational resources planned for expected protest volume and the political cost of expected participant harm. This distinction produces the crowd-management mechanism studied below.

The success-contingent reward also connects the model to work on selective incentives

and democratic participation. Aytaç and Stokes (2019) explain participation through costs that citizens associate with abstention, including emotional responses and social obligations. Wood's (2003) pleasure-in-agency similarly locates a reward in contributing rather than in passively receiving a public good. These approaches distinguish participation-contingent motivation from the policy benefit shared by participants and nonparticipants. The distinction supplies the strategic complementarity in the benchmark and clarifies the conditional role of the citizens' policy prize.

Little (2016) distinguishes two coordination problems that potential protesters face. The first is "political coordination," deciding whether to participate without knowing how many others are dissatisfied enough to join. The second is "tactical coordination," deciding when, where, and how to protest. His political coordination model also has a continuum of citizens with heterogeneous assessments and equilibria in cutoff strategies. He shows that better tactical information "almost always lowers the effective cost of protest, leading to larger (and more cohesive) antiregime action" (p. 152).

3 Modeling protests, repression, and policy change.

In this section, we develop a model of citizens who decide to protest to change a disliked policy. The citizens face a coordination and collective action problem. The protest changes the policy only if citizens' participation reaches a given threshold. The threshold is an uncertain fundamental, and citizens receive noisy private signals of it. On the other side, the state would prefer not to change the policy. To influence participation, it can set the expected repression cost borne by a participant. A realized mass protest consumes operational resources, while a policy that imposes greater expected harm invites legal, electoral, and reputational backlash. We begin with the citizens' participation problem and its complete-information benchmark. Next, we solve the global game. Then, we propose the two-stage sequential game in which the state sets expected participant cost

first, and we inspect the strategic implications of the model.

Table 1: Notation and role in the model

Object	Meaning and units	Role
τ	Critical participating share required for policy change	Realized fundamental
N^*	Share of citizens who participate	Equilibrium outcome
B_p	Public policy benefit from successful protest	Citizen payoff primitive
d_p	Success-contingent participation reward	Citizen payoff primitive
q_r	Probability or share of participants targeted	Measured component of the directive
s_r	Harm conditional on being targeted	Measured component of the directive
$c_r = q_r s_r$	Expected repression cost per participant	State choice
B_s	Government value of preserving the policy status quo	State payoff primitive
$\bar{N}(c_r)$	Expected share of citizens who participate	Anticipated outcome
$H(\bar{N}, c_r; \theta_0)$	Planned operational cost for expected protest volume and the chosen repression policy	State cost technology
$K(c_r; \theta_1)$	Political cost of expected participant harm	State cost technology
σ	Noise in citizens' private signals	Information parameter

3.1 The citizens' participation problem.

There is a continuum of citizens of mass one, indexed by $i \in [0, 1]$. Each citizen chooses between participating in the protest (P) or not (NP). Let $n_i \in \{0, 1\}$ indicate participation, and let $N^* \in [0, 1]$ denote the realized mass of participants. Protesters want to change a policy so gridlocked that ordinary representative channels struggle to move it. The protest is successful if participation reaches a critical threshold τ , so the policy changes if and only if $N^* \geq \tau$.

The payoff primitives are as follows. A change in the policy provides a collective and indivisible benefit B_p to all individuals in the society (i.e., a public good). Individuals who participate may get an expressive or emotional benefit d_p . They may feel, for example, that fighting for a just cause is good in and of itself. Participants also pay c_r , the expected repression cost borne by a participant. This object combines the probability of victimization with the conditional harm suffered if victimized. If q_r denotes reach and s_r conditional severity, then $c_r = q_r s_r$.

Assumption 1 (One-dimensional directive). The executive chooses from an institution-specific, one-dimensional menu of policing directives indexed by $c_r \in [0, \infty)$. The menu assigns a single policy package to each index value, with reach $q_r : [0, \infty) \rightarrow [0, 1]$, conditional severity $s_r : [0, \infty) \rightarrow [0, \infty)$, and $q_r(c_r)s_r(c_r) = c_r$.

The directive is the policy unit; q_r and s_r are its measurable components. This restriction fits institutions in which rules of engagement bundle deployment, targeting, and authorized force into stable policy packages. The menu retains the cost-minimizing package whenever an institution makes several packages with the same product feasible, and it absorbs the minimized cost into H . The functions H and K below evaluate costs along that feasible policy menu. Expected participant harm is therefore sufficient for the citizen's deterrence decision and indexes the packages compared by the executive. Conditional severity refers only to s_r .

Two features of the continuum setting must be made explicit, because they discipline how the primitives enter the participation margin.

First, each individual citizen is nonpivotal. With finitely many players, Δp_i would place the public good benefit B_p in the individual calculus. That term is the change in the probability of meeting the threshold induced by one's own participation. In the continuum, a single citizen has measure zero, so $\Delta p_i = 0$. The citizen receives B_p when the protest succeeds whether she participated or abstained, and B_p cancels from that comparison. The benchmark therefore isolates the participation-contingent reward as the motive of the

marginal participant.

Second, the expressive benefit must be success-contingent for participation to respond to the prospects of success at all. A participant enjoys the expressive reward d_p only if the movement succeeds. That satisfaction arrives only when the change actually happens. This success-contingency is a substantive assumption, and it generates the coordination problem.¹ Were d_p unconditional, participation would reduce to the comparison of d_p against c_r , independent of any belief about others, and the strategic structure would disappear.

With these two features, the net payoff of participating relative to abstaining, for citizen i , is

$$u_i(P) - u_i(NP) = \mathbb{1}\{N^* \geq \tau\} \cdot d_p - c_r,$$

since $B_p \cdot \mathbb{1}\{N^* \geq \tau\}$ enters both sides and cancels. A citizen therefore participates when her belief that the protest will succeed, times the expressive reward, exceeds the expected repression cost.

3.2 The complete-information benchmark.

Suppose first, as a benchmark, that τ is common knowledge. Assume $0 < \tau \leq 1$, the interesting range, and $c_r < d_p$, so that participation is worthwhile when success is assured. Then two pure strategy equilibria coexist, and the small-noise limit selects a unique interior prediction (Carlsson and van Damme 1993; Morris and Shin 1998, 2003). If every citizen participates, $N^* = 1 \geq \tau$, the protest succeeds, and each participant receives $d_p - c_r > 0$. No citizen gains by deviating, since abstaining forfeits $d_p - c_r$ while leaving success (and hence B_p) unchanged. If no citizen participates, $N^* = 0 < \tau$, the protest fails, and a lone deviator would pay c_r for a failed protest. Universal abstention is therefore also an equilibrium. If instead $c_r > d_p$, universal abstention is the unique equilibrium, because

1. See Bueno de Mesquita and Shadmehr (2023) and Wood (2003).

participation is dominated even under assured success.

The complete-information benchmark presents two problems.² The multiplicity makes prediction difficult, and the equilibrium set is insensitive to the parameters, including the repression policy itself. One could respond by selecting an equilibrium, for instance the mixed strategy equilibrium, and reading its participation probabilities as frequencies. We instead take seriously that the critical threshold is a variable citizens cannot observe. The size a protest must reach to force a policy change is written nowhere, unlike the threshold to win an election under plurality rule, which is common knowledge. How citizens form beliefs about τ is therefore central to any behavioral prediction.

3.3 The global game.

Let the critical threshold τ be drawn from the common prior $U[0, 1]$ and remain unobserved by citizens. Citizens and the state use this same proper prior. Each citizen receives a noisy private signal

$$x_i = \tau + \sigma \varepsilon_i,$$

where the ε_i are i.i.d. with a log-concave density f and c.d.f. F with full support on $\mathbb{R}$, independent of τ , and $\sigma > 0$ scales the noise. A citizen with a low signal believes the required crowd is small and success likely. A citizen with a high signal believes success is out of reach.

A strategy is a mapping from signals to participation. The participation payoff is decreasing in the (believed) threshold, so cutoff strategies are the natural place to look for equilibria. A citizen participates if and only if $x_i < \hat{x}$. Given such a cutoff, the mass of participants at true threshold τ is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

2. The result mirrors Bueno de Mesquita and Shadmehr (2023, Proposition 1).

which is continuous and strictly decreasing in τ . The protest therefore succeeds if and only if τ is below a unique critical value $\hat{\tau}$. The equilibrium is characterized by two conditions:³

$$\text{(belief consistency / critical mass):} \quad F\left(\frac{\hat{x} - \hat{\tau}}{\sigma}\right) = \hat{\tau},$$

$$\text{(indifference of the marginal citizen):} \quad \Pr(\tau \leq \hat{\tau} \mid x_i = \hat{x}) \cdot d_p = c_r.$$

The first condition says that at the critical threshold the crowd is exactly critical. The second says that a citizen whose signal equals the cutoff is indifferent between participating and abstaining. Under the proper prior, the marginal citizen's exact posterior success probability is

$$\pi_\sigma(\hat{x}, \hat{\tau}) = \frac{F(\hat{x}/\sigma) - F((\hat{x} - \hat{\tau})/\sigma)}{F(\hat{x}/\sigma) - F((\hat{x} - 1)/\sigma)}.$$

Thus a positive-noise monotone cutoff equilibrium solves

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \quad d_p \pi_\sigma(\hat{x}_\sigma, \hat{\tau}_\sigma) = c_r. \quad (1)$$

These equations retain the boundary terms generated by $U[0, 1]$. At an interior threshold they converge, as $\sigma \rightarrow 0$, to $(1 - \hat{\tau})d_p = c_r$. The limiting success threshold is therefore

$$\hat{\tau}(c_r) = \max\left\{1 - \frac{c_r}{d_p}, 0\right\}. \quad (2)$$

Normalized expected participant harm c_r/d_p determines the limiting threshold.⁴ Proposition 1 states the proper-prior convergence result.

Proposition 1 (Proper-prior small-noise selection). *Fix a policy with $\rho \equiv c_r/d_p \in (0, 1)$ and let $\hat{\tau}_0 = 1 - \rho$. For every $\delta \in (0, \min\{\hat{\tau}_0, 1 - \hat{\tau}_0\})$, there is a $\bar{\sigma} > 0$ such that equation (1) has*

3. These conditions correspond to the canonical regime-change game in Bueno de Mesquita and Shadmehr (2023, equation 1).

4. Equation (2) is the analog of $\theta^p = 1 - c$ in Bueno de Mesquita and Shadmehr (2023, equation 2 and Proposition 2), with c_r/d_p in the role of c .

an interior monotone cutoff equilibrium with $\hat{\tau}_\sigma \in [\hat{\tau}_0 - \delta, \hat{\tau}_0 + \delta]$ for every $\sigma \in (0, \bar{\sigma})$. Every selection from these local equilibria satisfies

$$\hat{x}_\sigma \longrightarrow \hat{\tau}, \quad \hat{\tau}_\sigma \longrightarrow \hat{\tau} = 1 - \frac{c_r}{d_p}.$$

The interior monotone limiting selection is therefore unique. If $c_r > d_p$, participation is strictly dominated. At $c_r = d_p$, full deterrence is the continuous extension of the interior small-noise selection, although other weak equilibria can coexist at equality. At $c_r = 0$, full participation is the corresponding limiting selection.

Online Appendix A.3 (p. A-4) gives the full proof, which uses the posterior lemmas of Online Appendix A.2 (p. A-2). Three remarks.

First, the selected interior branch exists for all sufficiently small positive noise and remains in a fixed compact neighborhood of $1 - c_r/d_p$. The bounded proper prior can also generate boundary-adjacent roots at positive noise. The theory follows the local branch that continues the interior global-games prediction. In the limiting game, protest succeeds if and only if $\tau \leq \hat{\tau}(c_r)$, so the state's comparative statics use its unique small-noise limit.

Second, the limiting outcome changes smoothly with the repression policy. The deterrence derivative is⁵

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{1}{d_p}.$$

Each unit of expected participant cost shrinks the set of fundamentals at which the protest succeeds at a constant rate, inversely proportional to the expressive reward. Repression is blunt against a strongly motivated citizenry (high d_p), and sharp against a weakly motivated one. This is the global games rendering of a familiar intuition: intrinsic

5. The notation $\partial \hat{\tau} / \partial c_r$ denotes the change in the limiting equilibrium success threshold induced by a marginal change in expected participant cost, with d_p fixed. The expression applies on the interior region $0 < c_r < d_p$; at the clamped boundary the relevant one-sided derivative is zero.

motivation erodes the deterrent effect of expected participant harm.

Third, the direct deterrence margin is governed by d_p alone. The zero partial effect of B_p is a benchmark result conditional on three assumptions: citizens form a continuum, the policy prize is public and available equally to participants and abstainers, and policy stakes are separable from success-contingent motivation. A larger policy stake may still raise d_p by intensifying grievance, identity, efficacy, or pleasure-in-agency, and it may raise the government's B_s . The model isolates the direct pivotality channel by fixing those pathways.

The state's problem is evaluated in the small-noise limit $\sigma \rightarrow 0$.⁶ The cutoff $\hat{x}$ then converges to $\hat{\tau}$ and participation becomes sharp: $N^*(\tau) \rightarrow 1$ for $\tau < \hat{\tau}$ and $N^*(\tau) \rightarrow 0$ for $\tau > \hat{\tau}$. Expected protest mass is therefore $\bar{N}(c_r) \equiv \mathbb{E}_\tau[N^*(\tau) \mid c_r] = \hat{\tau}(c_r)$. The limiting environment captures the extensive margin between a mass protest and no mobilization and keeps the state's planning problem transparent.

3.4 The state's problem.

The elected executive knows the policy dispute and its stakes, then fixes an operational directive before citizens receive their private information and mobilize. Police implement that directive during the episode.

Assumption 2 (Within-episode commitment). The state chooses c_r after the policy dispute is known and before citizen mobilization, and c_r remains fixed during the protest episode.

The timing captures advance deployment, rules of engagement, authorized tactics, and expectations formed from directives and prior practice. The directive is publicly

6. The executive's optimization problem below is defined over the limiting policy mapping $c_r \mapsto \hat{\tau}(c_r)$. The positive-noise calculations in Online Appendix A.4 (p. A-6) illustrate the robustness of the selected citizen branch; they do not supply a separate convergence theorem for a finite-noise state optimizer.

announced, legally recorded, or credible under standing practice before citizens mobilize. Legal review, administrative compliance, press scrutiny, and anticipatory electoral response can therefore impose a political cost when the directive is authorized, even if deterrence prevents its use.

Let $H(\tilde{N}, c_r; \theta_0)$ denote the operational cost of preparing for expected protest mass $\tilde{N}$ under policy c_r . It includes personnel, equipment, transport, command capacity, and other resources committed in anticipation of protest, including preparedness expenditures incurred before turnout is realized.

Assumption 3 (Planned operational cost). The function $H(\tilde{N}, c_r; \theta_0)$ is twice continuously differentiable on $\tilde{N} \in [0, 1]$ and $c_r \geq 0$, satisfies $H(0, 0; \theta_0) = 0$, $H_N > 0$, $H_c \geq 0$, and $H_{N\theta_0} > 0$.

The shifter θ_0 raises the marginal cost of preparing for additional protesters. The cross-partial $H_{c\theta_0}$ permits the same shifter to change the direct marginal operational cost of harsher tactics.⁷

The same subscript convention applies to the political-cost function.⁸

Assumption 4 (Political cost of the authorized directive). The function $K(c_r; \theta_1)$ is twice continuously differentiable on $c_r \geq 0$, satisfies $K(0; \theta_1) = 0$, $K_c(0; \theta_1) = 0$, $K_c(c_r; \theta_1) > 0$ for $c_r > 0$, $K_{cc}(c_r; \theta_1) > 0$, $K_{c\theta_1}(c_r; \theta_1) > 0$, and $K_c(c_r; \theta_1) \rightarrow \infty$ as $c_r \rightarrow \infty$.

Strict convexity captures rising marginal legal, electoral, administrative, and reputational costs of authorizing greater expected harm per participant. The cross-partial gives θ_1 a transparent institutional interpretation among democracies: a larger θ_1 raises the marginal political price of every positive c_r .

7. Subscripts on H denote partial derivatives evaluated at $(\tilde{N}, c_r; \theta_0)$. Thus $H_N \equiv \partial H / \partial \tilde{N}$, $H_c \equiv \partial H / \partial c_r$, $H_{NN} \equiv \partial^2 H / \partial \tilde{N}^2$, $H_{Nc} \equiv \partial^2 H / (\partial \tilde{N} \partial c_r)$, $H_{cc} \equiv \partial^2 H / \partial c_r^2$, $H_{N\theta_0} \equiv \partial^2 H / (\partial \tilde{N} \partial \theta_0)$, and $H_{c\theta_0} \equiv \partial^2 H / (\partial c_r \partial \theta_0)$. Every partial derivative fixes the function's remaining arguments.

8. Thus $K_c \equiv \partial K(c_r; \theta_1) / \partial c_r$, $K_{cc} \equiv \partial^2 K(c_r; \theta_1) / \partial c_r^2$, and $K_{c\theta_1} \equiv \partial^2 K(c_r; \theta_1) / (\partial c_r \partial \theta_1)$. The function's other argument remains fixed in each derivative.

The state keeps $B_s > 0$ when the status quo survives; its policy payoff after protest-induced change is normalized to zero. With the proper planning distribution $\tau \sim U[0, 1]$, the success probability and expected protest mass in the small-noise limit both equal $\hat{\tau}(c_r)$. Expected utility is

$$u_S(c_r) = (1 - \hat{\tau}(c_r))B_s - H(\hat{\tau}(c_r), c_r; \theta_0) - K(c_r; \theta_1). \quad (3)$$

For $c_r \in (0, d_p)$, the marginal payoff is⁹

$$u_c(c_r) = \frac{B_s + H_N}{d_p} - H_c - K_c. \quad (4)$$

The term $(B_s + H_N)/d_p$ values the status quo and planned operational resources saved when deterrence reduces expected protest mass. The term H_c is the direct marginal operational cost of implementing harsher tactics. The term K_c is the marginal political cost of expected participant harm.

Assumption 5 (Interior curvature). The objective is twice continuously differentiable and strictly concave on $(0, d_p)$, with $u_{cc}(c_r) < 0$ and right derivative $u_c(0^+) > 0$.

Using equation (4), strict concavity is equivalent to

$$\frac{H_{NN}}{d_p^2} - \frac{2H_{Nc}}{d_p} + H_{cc} + K_{cc} > 0.$$

This condition allows planned operational cost to be nonlinear in expected protest mass and expected participant harm while keeping a unique policy response.

Proposition 2 (The optimal repression policy). *Under Assumptions 2–5, the state has a unique optimal policy on $[0, d_p]$. The solution is interior and satisfies $u_c(c_r^*) = 0$ when the left*

9. Here $u_c \equiv \partial u_S / \partial c_r$ and $u_{cc} \equiv \partial^2 u_S / \partial c_r^2$. These derivatives vary the state's policy c_r and allow the induced expected mass $\bar{N}(c_r) = \hat{\tau}(c_r)$ to adjust, while fixing B_s , d_p , θ_0 , θ_1 , and the functional forms of H and K .

derivative $u_c(d_p^-) < 0$; otherwise the state chooses the full-deterrence corner $c_r^* = d_p$. Expected harm above d_p is dominated because the success threshold and expected protest mass are already clamped at zero while H and K are weakly increasing in c_r .

Online Appendix A.5 (p. A-7) gives the proof.

3.5 Comparative statics.

The general technology separates effects that follow directly from the model from the condition governing the crowd-cost mechanism.

Proposition 3 (Comparative statics). *At the interior solution,*¹⁰

$$\begin{aligned} \frac{\partial c_r^*}{\partial B_s} &= -\frac{1/d_p}{u_{cc}} > 0, & \frac{\partial c_r^*}{\partial \theta_1} &= \frac{K_{c\theta_1}}{u_{cc}} < 0, \\ \frac{\partial c_r^*}{\partial d_p} &= \frac{B_s + H_N + c_r^* H_{Nc} - (c_r^*/d_p) H_{NN}}{d_p^2 u_{cc}}, & \frac{\partial c_r^*}{\partial B_p} &= 0, \end{aligned}$$

Thus $\partial c_r^*/\partial d_p < 0$ when $B_s + H_N + c_r^* H_{Nc} > (c_r^*/d_p) H_{NN}$. For the crowd-cost shifter,

$$\frac{\partial c_r^*}{\partial \theta_0} = -\frac{H_{N\theta_0}/d_p - H_{c\theta_0}}{u_{cc}}. \quad (5)$$

Hence a higher marginal cost of preparing for protesters increases expected participant harm if and only if

$$\frac{H_{N\theta_0}}{d_p} > H_{c\theta_0}. \quad (\text{Crowd-cost condition})$$

10. Each expression $\partial c_r^*/\partial z$ or $\partial \hat{\tau}^*/\partial z$ is an equilibrium comparative-static derivative. It allows the optimizing policy, expected protest mass, and success threshold to adjust to primitive z while the other listed primitives and both cost-function families remain fixed. All derivatives of H on the right-hand side are evaluated at $(\bar{N}, c_r; \theta_0) = (\hat{\tau}^*, c_r^*; \theta_0)$; derivatives of K are evaluated at $(c_r^*; \theta_1)$.

A rise in B_s lowers the equilibrium success probability, whereas a rise in θ_1 raises it. When the crowd-cost condition holds, a rise in θ_0 lowers the success probability. A rise in d_p raises it whenever the stated condition for $\partial c_r^*/\partial d_p < 0$ holds.

Online Appendix A.6 (p. A-8) gives the proof. The crowd-cost condition is the paper's central technological result. The term $H_{N\theta_0}/d_p$ is the additional planning cost saved when deterrence reduces expected protest mass. The term $H_{c\theta_0}$ captures how the same shifter changes the direct marginal operational cost of harsher tactics. The first force raises the state's optimal expected harm per participant; the second tempers that increase and can reverse it.

Two organizational technologies give the condition substantive content. Let w denote the price of personnel and logistical inputs, and let $A(c_r)$ collect operational costs that do not vary with w . In a *deployment-separable* system,

$$H^D(\bar{N}, c_r; w) = w\bar{N} + A(c_r). \quad (6)$$

The input price raises the marginal cost of preparing for an additional protester, $H_{Nw}^D = 1$, without changing the direct marginal cost of the directive, $H_{cw}^D = 0$. Thus $\partial c_r^*/\partial w > 0$: higher personnel, transport, or crowd-control deployment costs make deterrence more valuable.¹¹

In a *tactic-intensive* system, the same inputs also implement the authorized level of expected harm:

$$H^T(\bar{N}, c_r; w) = w(\bar{N} + \lambda c_r) + A(c_r), \quad \lambda > 0. \quad (7)$$

Now $H_{Nw}^T = 1$ and $H_{cw}^T = \lambda$. The price increase raises expected participant harm when

11. The derivative $\partial c_r^*/\partial w$ allows the optimal directive and anticipated protest mass to adjust to the input price w ; B_s , d_p , θ_1 , and the functional forms of A and K remain fixed. The other arguments of H remain fixed in the cross-partial H_{Nw} and H_{cw} .

$\lambda < 1/d_p$ and lowers it when $\lambda > 1/d_p$, as Figure 2 shows.¹² The second case describes organizations in which specialized officers, less-lethal munitions, medical support, detention processing, or documentation requirements make stronger directives especially input-intensive. Equations (6) and (7) yield a discriminating implication: comparable input-price increases lead the state to authorize more expected participant harm where they load mainly on personnel deployment and less expected participant harm where they load heavily on implementing force.

The institutional comparison follows directly from $K_{c\theta_1} > 0$. Environments with stronger legal liability, judicial oversight, electoral accountability, or media scrutiny raise the marginal political price of expected harm and therefore reduce c_r^* . This gives the democratic setting a formal role through the cost schedule faced by the elected executive.

For the transparent benchmark, use

$$H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}, \quad K(c_r; \theta_1) = \theta_1 c_r^2.$$

This is the deployment-separable technology with $A(c_r) = 0$. Then $H_{N\theta_0} = 1$, $H_{c\theta_0} = 0$, the crowd-cost condition holds strictly, and the solution reduces to

$$c_r^* = \min \left\{ \frac{B_s + \theta_0}{2\theta_1 d_p}, d_p \right\}, \quad \hat{t}^* = \max \left\{ 1 - \frac{B_s + \theta_0}{2\theta_1 d_p^2}, 0 \right\}.$$

Figure 1 uses this additive-quadratic specialization to show the mechanism.

In the benchmark, a higher θ_0 raises expected participant harm. It increases the planned resources saved by reducing expected protest mass without changing the direct

12. The loading λ converts one unit of expected participant harm into the same operational-input units used to measure $\bar{N}$; its units are therefore units of anticipated participation per unit of harm. Because d_p and c_r share payoff units, $1/d_p$ has the same units as λ . The numerical cutoff is tied to those normalizations and is not directly portable across differently scaled data.

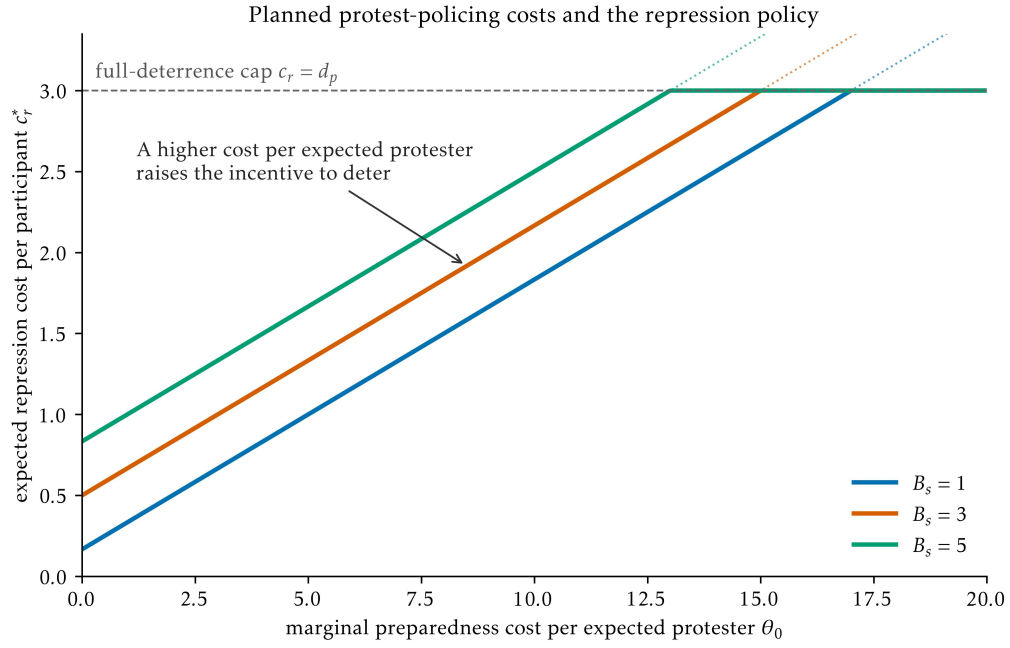


Figure 1: The crowd-management mechanism in the additive-quadratic benchmark, $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$ and $K(c_r; \theta_1) = \theta_1 c_r^2$, with $\theta_1 = 1$ and $d_p = 3$. Solid lines show the effective policy for three values of the government's status quo stake; dotted lines show the unconstrained interior solution. A higher marginal cost of preparing for protesters raises expected participant harm until the full-deterrence cap binds.

marginal operational cost of harsher tactics. A higher θ_1 lowers c_r by steepening the marginal political-cost schedule. Greater intrinsic motivation d_p lowers the interior policy because motivated citizens are harder to deter. At the full-deterrence corner, c_r tracks d_p one for one. Online Appendix A.1 (p. A-2) maps the policy regions of this benchmark and plots its comparative statics in B_s and d_p .

The benchmark is the case $\lambda = 0$ with $w = \theta_0$, in which a higher input price leaves the direct cost of harsher tactics unchanged and therefore always raises expected harm. Figure 2 adds the technology in equation (7) with $A(c_r) = 0$ and the same political cost $K(c_r; \theta_1) = \theta_1 c_r^2$. Solving the state's problem then gives the interior policy and its response to the input price,

$$c_r^* = \frac{B_s + w(1 - \lambda d_p)}{2\theta_1 d_p}, \quad \frac{\partial c_r^*}{\partial w} = \frac{1 - \lambda d_p}{2\theta_1 d_p}.$$

Panel A fixes $B_s = 5$ and $d_p = 3$ and raises the input price w under four loadings λ . All four lines start at $B_s/(2\theta_1 d_p)$, and the line for $\lambda = 0$ reproduces the $B_s = 5$ line of Figure 1. The same price increase raises expected participant harm when $\lambda < 1/d_p$, leaves it unchanged when $\lambda = 1/d_p$, and lowers it when $\lambda > 1/d_p$. Panel B maps the sign of that response over d_p and λ . The boundary between its two regions depends on neither B_s nor θ_1 , since B_s enters only the level of the policy and θ_1 scales its response to w without changing its sign.

Two features of these results deserve emphasis.

First, the status quo prize B_s has an unambiguously positive effect at every interior optimum. The monotone cutoff equilibrium makes the deterrence technology the linear relation (2), so the marginal value of expected participant harm rises with the government's stake in the disputed policy.

Second, the direct partial effect of B_p equals zero under the benchmark's nonpivotality and separability assumptions. Policy stakes affect repression through the participation reward d_p and the government's status quo value B_s . The relevant distinction is between the prize's direct public-good component and the motivational and governmental channels through which politically important disputes alter behavior.

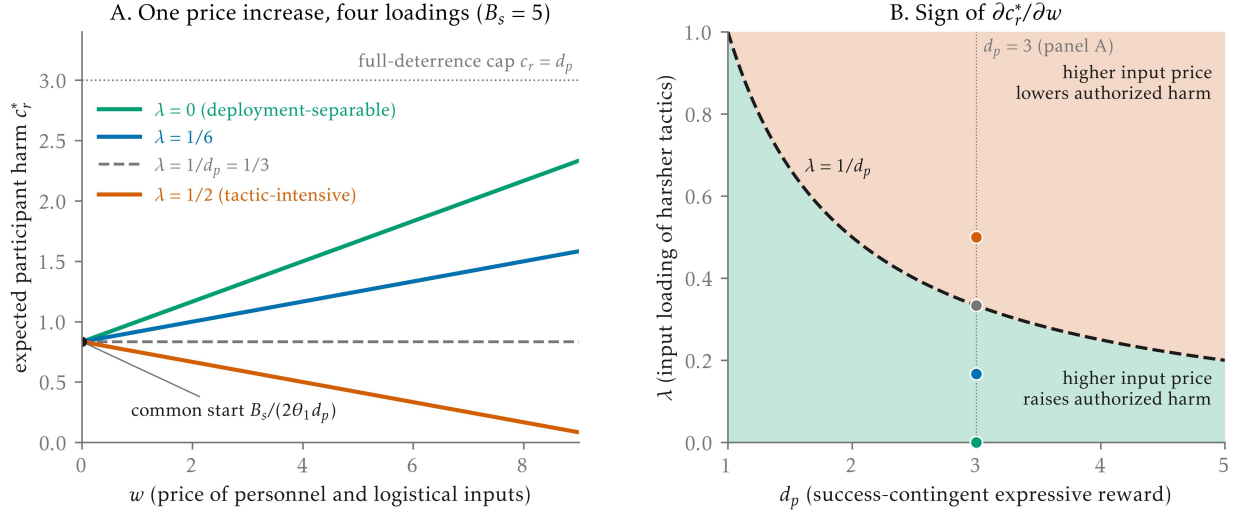


Figure 2: Reversal of the budget effect when the same inputs implement force. Both panels use the technology $H^T(\tilde{N}, c_r; w) = w(\tilde{N} + \lambda c_r)$ and the political cost $K(c_r; \theta_1) = \theta_1 c_r^2$, with $\theta_1 = 1$. Panel A fixes $B_s = 5$ and $d_p = 3$ and varies the input price w for four loadings λ , and every policy shown is interior. Panel B shows the sign of $\partial c_r^*/\partial w$ over d_p and λ , with the boundary $\lambda = 1/d_p$ dashed and the four loadings of panel A marked at $d_p = 3$.

3.6 Directive costs and exposure-dependent political costs.

The baseline $K(c_r; \theta_1)$ is paid when the executive authorizes an observable directive. Democratic institutions may also impose consequences after citizens are exposed to it. In the small-noise limit, total expected harm in the population is

$$T(c_r) = c_r \hat{\tau}(c_r).$$

This aggregation uses the benchmark's homogeneous expected harm per participant and sharp small-noise participation.¹³ Let $L(T; \theta_2)$ be an increasing, convex political cost of

13. At positive noise with homogeneous exposure, the corresponding object is $c_r \mathbb{E}_\tau[N^*(\tau)]$. With heterogeneous exposure c_{ri} , it is $\mathbb{E}_\tau[\int_0^1 n_i(\tau) c_{ri} di]$. Casualty, injury, or detention totals therefore map to T only after accounting for turnout and the distribution of exposure among participants.

exposure, in addition to the directive cost K . The state then maximizes

$$u_S^E(c_r) = (1 - \hat{\tau})B_s - H(\hat{\tau}, c_r; \theta_0) - K(c_r; \theta_1) - L(c_r \hat{\tau}; \theta_2).$$

At an interior policy its marginal payoff is¹⁴

$$u_c^E(c_r) = \frac{B_s + H_N}{d_p} - H_c - K_c - L_T \left(\hat{\tau} - \frac{c_r}{d_p} \right). \quad (8)$$

Exposure-dependent accountability changes the optimal level because total expected harm first rises and then falls as deterrence reduces the crowd. Differentiating equation (8) in the planning-cost shifter again gives $H_{N\theta_0}/d_p - H_{c\theta_0}$ when the extended objective remains strictly concave and L does not depend on θ_0 . The crowd-cost condition therefore continues to determine the sign of $\partial c_r^*/\partial \theta_0$. The directive cost remains positive at full deterrence, while the exposure cost reaches zero with expected turnout.

3.7 An extension: Repression, protests, and emotions.

The benchmark holds success-contingent motivation fixed. To isolate one feedback, let citizens observe the directive c_r before choosing whether to participate and let $d_p(c_r) = d_0 + e(c_r)$. The observable object can be a public rule, announced tactic, or credible expectation formed from prior police practice. The threshold becomes

$$\hat{\tau}(c_r) = \max \left\{ 1 - \frac{c_r}{d_0 + e(c_r)}, 0 \right\}.$$

14. Here $L_T \equiv \partial L(T; \theta_2)/\partial T$, evaluated at $T = c_r \hat{\tau}(c_r)$. The derivative varies c_r , allows $\hat{\tau}(c_r)$ and total expected harm to adjust, and fixes B_s , d_p , θ_0 , θ_1 , θ_2 , and the cost-function families. Since $\partial \hat{\tau}/\partial c_r = -1/d_p$, $\partial(c_r \hat{\tau})/\partial c_r = \hat{\tau} - c_r/d_p$.

At an interior threshold, differentiation gives¹⁵

$$\frac{\partial \hat{\tau}}{\partial c_r} = - \frac{d_0 + e(c_r) - c_r e'(c_r)}{(d_0 + e(c_r))^2}.$$

Define the elasticity of success-contingent motivation to expected participant harm as $\varepsilon(c_r) = c_r e'(c_r)/d_p(c_r)$.

Proposition 4 (Deterrence and backlash). *Expected participant harm deters, $\partial \hat{\tau}/\partial c_r < 0$, if and only if $\varepsilon(c_r) < 1$; it backfires, $\partial \hat{\tau}/\partial c_r > 0$, if and only if $\varepsilon(c_r) > 1$.*

Online Appendix A.7 (p. A-9) gives the proof. The extension converts the punishment puzzle into a scale-free condition. Anticipated repression deters when motivation rises less than proportionally with expected harm and recruits when it rises more than proportionally.

The emotional response also changes the directive the state is willing to choose. Index the strength of that response by $\alpha \geq 0$ and write $d_p(c_r; \alpha) = d_0 + \alpha e(c_r)$, so $\alpha = 1$ recovers the preceding specification. At an interior threshold, define the marginal deterrent return

$$D(c_r, \alpha) \equiv - \frac{\partial \hat{\tau}(c_r; \alpha)}{\partial c_r} = \frac{d_p(c_r; \alpha) - c_r \alpha e'(c_r)}{d_p(c_r; \alpha)^2} = \frac{1 - \varepsilon(c_r; \alpha)}{d_p(c_r; \alpha)}, \quad (9)$$

where $\varepsilon(c_r; \alpha) = c_r \alpha e'(c_r)/d_p(c_r; \alpha)$.¹⁶

Proposition 5 (Emotions and the state's policy). *Suppose $e(c_r) \geq 0$ and $e'(c_r) \geq 0$, and let $c_r^*(\alpha) > 0$ be an interior optimum with $\hat{\tau}(c_r^*; \alpha) \in (0, 1)$ at which the state's objective is strictly*

15. The derivative $\partial \hat{\tau}/\partial c_r$ varies the observable policy c_r and fixes d_0 and the response function $e(\cdot)$; $e'(c_r)$ is the ordinary derivative of that function evaluated at c_r .

16. The partial derivative $\partial \hat{\tau}(c_r; \alpha)/\partial c_r$ varies the directive while fixing d_0 , α , and the response function $e(\cdot)$. In the proof, $D_\alpha \equiv \partial D/\partial \alpha$, $G_\alpha \equiv \partial G/\partial \alpha$, and $G_{c_r} \equiv \partial G/\partial c_r$ hold the other displayed argument fixed. The derivative $\partial c_r^*/\partial \alpha$ varies the strength of the emotional response while allowing the state to reoptimize; B_s , w , θ_1 , $A(\cdot)$, $K(\cdot)$, and $e(\cdot)$ remain fixed.

concave. First, any such optimum lies in the deterrence region: $\varepsilon(c_r^*; \alpha) < 1$. Second, under the deployment-separable technology $H^D(\bar{N}, c_r; w) = w\bar{N} + A(c_r)$, a stronger nonzero emotional response reduces the state's chosen expected harm:

$$\frac{\partial c_r^*}{\partial \alpha} < 0 \quad \text{if} \quad e(c_r^*) + c_r^* e'(c_r^*) > 0.$$

Online Appendix A.8 (p. A-10) gives the proof. A state does not choose a locally backfiring directive at an interior optimum: such a directive increases expected protest while also carrying operational and political costs. Even before backfire occurs, stronger emotional responsiveness weakens the marginal deterrent return. In a deployment-separable system, the state responds by selecting less expected harm. Evidence that mobilization follows a coercive tactic is informative about this mechanism when the tactic causally changes both expected participant harm and success-contingent motivation. The timing concerns an observable ex ante policy. Across repeated episodes, prior realized violence can determine citizens' current expectations of that policy.

4 Discussion: contribution, observables, and research design.

The theory distinguishes the operational cost of preparing for the expected protest volume from the political cost of authorizing expected harm per participant. These costs affect the state's choice differently. A higher marginal cost of preparing for protesters makes deterrence more valuable, whereas the marginal legal, electoral, administrative, and reputational costs of the directive restrain the policy used to deter. The derived crowd-cost condition determines the direction of the crowd-cost effect when the same cost environment also changes the direct marginal operational cost of harsher tactics. Expected participant harm rises only when the planning resources saved through deterrence exceed

that additional marginal operational cost. Equation (8) adds political consequences from total expected harm while preserving this sign condition.

A substantial literature connects fiscal consolidation to protest. A separate literature explains protest-policing tactics, mainly through threat perception, security diffusion, and legal change. Noakes and Gillham (2007) and Gillham (2011) attribute the turn toward more preemptive tactics to that perceived threat. Selmini and Di Ronco (2023, p. 206) describe the resulting tactics in detail: “the establishment of no-protest zones, the increased use of less-than-lethal weapons, the strategic use of arrests before protests, and reinvigoration of surveillance of perceived problematic social movements.” None of these three accounts discusses police budgets. To our knowledge, no study asks whether fiscal constraint on the policing apparatus changes the severity of authorized tactics. The model asks that question, and its central prediction remains untested. The remainder of this section reads existing evidence through the model.

The model assigns separate roles to three political quantities. The government’s B_s is the value of preserving the disputed status quo. Citizens’ d_p is the participation-contingent reward from contributing to a successful movement. The public policy benefit B_p has no direct pivotality effect in the continuum benchmark, but politically important disputes can affect both d_p and B_s . Klein and Regan’s (2018) cross-national protest-event analysis provides suggestive evidence about status quo stakes. They find that high concession costs, meaning demands that are expensive for the ruling coalition to accept, are associated with coercive state responses. This pattern is consistent with the positive effect of B_s . They also find that high disruption costs encourage accommodation. That result points in the opposite direction from the additive crowd-cost benchmark, but it concerns economic and public-order losses tied to a protest’s location, duration, and size. The model’s θ_0 instead changes the marginal operational cost of preparing for an additional participant. The contrast shows why the two costs should not be combined into a single measure of how expensive a protest is for the government.

The observable counterpart of c_r combines reach and intensity. Let q_r be the probability that a participant is targeted and s_r the harm conditional on targeting. An event-level measure of expected participant harm joins participant exposure to evidence about deaths, injuries, detention, sexual violence, psychological trauma, and other coercive tactics, so that $c_r = q_r s_r$. Under Assumption 1, observed directives trace an institution-specific policy menu. The executive does not choose q_r and s_r independently. Measuring both components constructs the deterrence index and reveals how police organizations package reach and conditional severity. Castro (2025) shows the value of tactic-specific information by coding civilian complaints during Chile’s 2019 protest cycle by tactic, municipality, and day and validating them against media reports and police ammunition records. She finds that selective high-severity tactics generated short-run mobilization and that low-severity tactics increased protest regardless of selectivity. Those results are consistent with the backfire region of Proposition 4 for the observed tactic combinations, although they do not estimate the model’s motivational elasticity.

Estimating the operational cost $H(\bar{N}, c_r; \theta_0)$ requires the government’s anticipated protest volume and the resources committed before an episode. The relevant inputs include scheduled personnel, overtime capacity, transport, equipment and consumables, temporary command infrastructure, detention capacity, and medical support. Benchmarking their prices against comparable labor and procurement costs measures opportunity cost across jurisdictions. Budget documents, deployment plans, procurement records, and after-action accounts can distinguish standing capacity from resources added in anticipation of protest. They can also estimate the loading λ in equation (7): input-price changes that mainly raise deployment cost predict greater expected participant harm, while price changes that strongly raise the cost of implementing force predict restraint. Within-jurisdiction changes are especially informative because ordinary crime also affects resource demand and police capacity.

In Lithuania, Juska and Woolfson (2012) record police allocations falling by 7.8 percent

in 2009, with a further cut proposed at 21.3 percent. The authors report that “real wages for rank-and-file officers had declined by almost one fifth.” The same period brought centralized command and tactics the authors characterize as “policing by threat of violence rather than policing by consent.” Centralization “allowed for the re-distribution and redirection of limited and shrinking police resources towards the strategic management of potential social unrest in the capital.” Their explanation differs from ours. They attribute it to doubts about whether officers who were themselves “victims of austerity cuts” would remain reliable, a channel distinct from deployment cost.

Officers in two further cases attribute violence to scarcity at the point of deployment. Greek riot police interviewed by Vasilaki (2016, p. 20) name numerical shortfall directly: “Operations are designed in a way that makes us violent. We are few against many.” British officers interviewed after the 2011 riots “said it was a lack of resources that resulted in them losing control,” and the Metropolitan Police afterward sought water cannon and authorized baton rounds (O’Sullivan 2017, pp. 35–36). Inspectors in England and Wales record commanders reporting that “the powers are sufficient; it is the ability to implement them that is the challenge due to lack of resources” (HMICFRS 2021, p. 97). The same inspection found mass arrest unavailable, because “the police do not have the capacity to make mass arrests in this scenario” (p. 97). These accounts record officers and commanders reasoning about severity under a resource constraint, which is the reasoning the model formalizes.

Fiscal pressure also produced restraint in these same episodes, which the crowd-cost condition anticipates. Vasilaki (2016) reports that Greek riot-squad numbers rose during austerity. O’Sullivan (2017, p. 198) finds the Metropolitan Police extending protest liaison partly for “the possibility it can present for reducing the amount of police resources that are mobilised at protests.” That extension lowered severity under the same budget pressure that produced the baton-round authorization. Whether a cost shock raises or lowers expected participant harm depends on how it loads across deployment and tactical

implementation. The crowd-cost condition (Crowd-cost condition) specifies which sign holds.

In Spain, police recruitment fell sharply after 2011, and enforcement shifted toward fines. Oliver Olmo and Urda Lozano (2015, p. 1322) name this pattern *burorrepresión* and describe it as “an invisible attack of coercion and intimidation.” Authorities adopt it because public violence against demonstrators arouses “a wave of sympathy and solidarity” (p. 1322). Oliver and Urda (2019, p. 84) add that “many people find it impossible to pay them in a time of crisis,” so a fixed nominal fine imposes greater expected cost as incomes fall. Administrative fines raise c_r while consuming little operational capacity, and they also generate revenue. In the notation of equation (7), such an instrument has a negative direct marginal operational cost. The Spanish literature attributes the shift to the political cost of visible force, which the model represents as θ_1 . The model identifies a second margin operating at the same time. This case does not show how to separate the two, and that remains an empirical question.

Institutions shift $K(c_r; \theta_1)$ by changing the consequences of authorizing a directive for elected officials. Published rules of engagement, judicial preclearance, legislative oversight, internal compliance review, and press access expose the authorization itself to scrutiny. Civil liability, casualty investigations, and electoral reactions to documented injuries instead load on $L(T; \theta_2)$. Aytaç, Schiumerini, and Stokes (2017) study backlash movements in Turkey, Brazil, and Ukraine and find that governments with stable electoral support were more willing to use harsh measures, whereas electorally insecure governments had stronger incentives to reduce violence. This pattern is consistent with the model’s claim that a lower marginal electoral cost permits greater expected participant harm. At a broader level, Davenport and Armstrong (2004) find that democracy reduces human-rights violations only above a threshold of democratic development. Their country-level outcome is not specific to protest directives, but the nonlinear result is consistent with interpreting θ_1 as meaningful institutional variation among democracies. Research designs can use the

timing of these institutions to distinguish changes in the marginal directive cost K_c from changes in the marginal exposure cost L_T .

Existing evidence also points to the motivational channel, while showing that participants need not respond uniformly. Adam-Troian, Çelebi, and Mahfud's (2020) cross-sectional study of 523 Yellow Vest protesters finds that exposure to police violence is positively associated with intentions to attend future demonstrations and to sacrifice for the movement. Perceived loss of significance and identification with the Yellow Vests mediate those associations. C. M. Sullivan and Davenport's (2017) panel study of the Republic of New Africa finds organizational backlash but different individual responses: members who personally experienced repression became more likely to participate, while members who were not directly repressed withdrew. These findings are consistent with a positive motivational response for some citizens, but they also caution against treating $e(c_r)$ as homogeneous. Survey and experimental data can measure d_p and the elasticity in Proposition 4 more directly. Items can measure political efficacy, anticipated satisfaction from contributing to collective success, and willingness to participate under stated probabilities of success and victimization. A factorial design can vary victimization reach, conditional harm, and the probability of policy success separately. The resulting responses distinguish the direct deterrent effect of expected harm from any change in success-contingent motivation.

Event records can first describe the joint variation in anticipated turnout, planned operational inputs, coercive reach, conditional harm, and institutional constraints. Suitable designs can then use institutional reforms and predetermined changes in labor or procurement costs to isolate particular margins. Two existing studies already report associations of this kind. H. Sullivan (2021, p. 587) finds that where Mexican subnational governments have bureaucratic capacity, "protest events are less likely to be managed using a repressive response." The crowd-cost condition would explain that association through the marginal cost of managing participants. The same study qualifies the accountability channel, re-

porting that “electoral competition is not a significant factor reducing the likelihood of repressive responses to protest” (H. Sullivan 2021, p. 587). Beck (2023) finds that United States municipalities funding one percent more of their budgets from fines and fees experienced four percent more police killings, which locates a fiscal determinant of coercive severity outside the protest setting. The comparative statics specify the expected pattern. Status quo stakes raise expected participant harm, and stronger marginal accountability lowers it. A planning-cost shifter raises expected harm only where the resources saved through deterrence exceed its effect on the marginal operational cost identified by the crowd-cost condition.

5 Conclusion.

This paper explains why a government that finds large protests expensive to manage may threaten harsher repression before people mobilize. The logic has two steps. Expected harm discourages some citizens from joining. A smaller expected crowd lets the government plan fewer officers, vehicles, detention spaces, and supplies. Harsher repression becomes attractive when the resources saved by preparing for a smaller crowd exceed the added operational and political costs of authorizing force. What matters is the cost of managing one more protester. The total price of maintaining a police force is a different quantity.

This tradeoff also explains why governments facing similar protests can respond differently. Consider an economic downturn that cuts tax revenue while raising demand for unemployment benefits, emergency transfers, and other social spending. The resulting fiscal pressure leaves less money for police overtime, recruitment, and equipment. If officials still expect a large protest, they may be unable to expand deployment as they otherwise would. Where revising the use-of-force directive is comparatively cheap, the executive may compensate by authorizing tactics that expose each participant to greater expected harm,

hoping that stronger deterrence allows a smaller force to manage the episode. The result still depends on how the police organization produces force. If stronger tactics require specialized officers, equipment, documentation, medical support, or detention capacity, the budget pressure can instead produce restraint. Legal review, political oversight, and electoral accountability further change the price that an elected government pays when it authorizes expected harm. Repression depends on which resources have become scarce and on how the directive uses them.

Citizens complete the argument because the crowd is not fixed in advance. Each person weighs the expected harm of participation against the satisfaction of helping a protest succeed. A stronger directive usually discourages participation and makes policy change less likely. Anger can reverse that result, but only when the added motivation grows faster than the expected harm. Repression can therefore anger citizens without producing a larger protest; anger and backfire are separate outcomes. When the state anticipates a stronger emotional response, it chooses less expected harm in the deployment-separable benchmark.

The argument points to a dynamic question for future research. Governments announce directives and plan deployments before an episode, but commanders and citizens also learn after turnout and injuries become visible. Comparing the initial directive with later tactical revisions would show when governments use an early threat to avoid a costly crowd, when operational demands force restraint, and when observed violence creates the mobilization that the directive was meant to prevent.

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Online Appendix

Repressive Democracies: State Repression, Protest, and Policy Change

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A.1 Policy regions and comparative statics for the additive-quadratic benchmark.

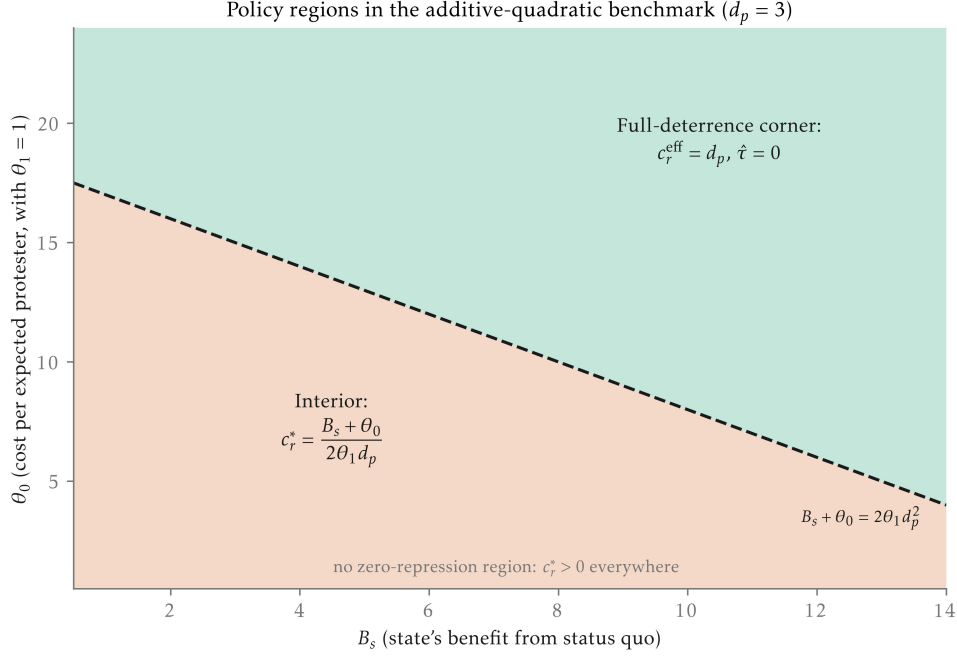


Figure A1: Interior and full-deterrence regions for $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$ and $K(c_r; \theta_1) = \theta_1 c_r^2$. The boundary is $B_s + \theta_0 = 2\theta_1 d_p^2$.

A.2 Posterior beliefs under the common proper prior.

Lemma 1 (Proper-prior posterior). Suppose $\tau \sim U[0, 1]$ and $x_i = \tau + \sigma \varepsilon_i$, where ε_i has density f and c.d.f. F . For $t \in [0, 1]$,

$$\Pr(\tau \leq t \mid x_i = x) = \frac{F(x/\sigma) - F((x-t)/\sigma)}{F(x/\sigma) - F((x-1)/\sigma)}.$$

Proof. The likelihood of x conditional on τ is $f((x-\tau)/\sigma)/\sigma$. The uniform prior restricts

Online Appendix: Repressive Democracies

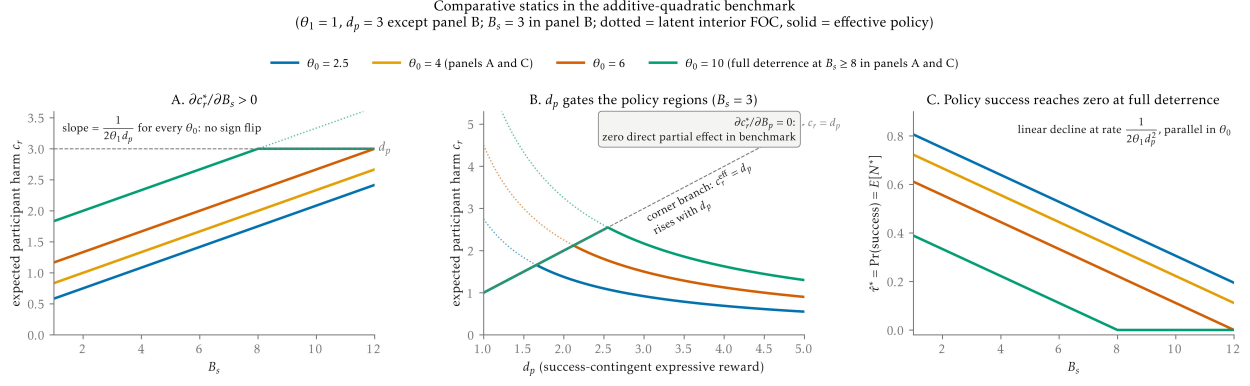


Figure A2: Region-conditional comparative statics for the additive-quadratic benchmark. Panel A varies the government's status quo stake. Panel B varies success-contingent motivation. Panel C shows the corresponding probability of policy change. Dotted curves are unconstrained interior solutions and solid curves are effective policies. The vertical axes label c_r as expected participant harm, not harm conditional on targeting.

its support to $[0, 1]$. Substituting $u = (x - \tau)/\sigma$ gives

$$\int_0^t \frac{1}{\sigma} f\left(\frac{x - \tau}{\sigma}\right) d\tau = F(x/\sigma) - F((x - t)/\sigma),$$

$$\int_0^1 \frac{1}{\sigma} f\left(\frac{x - \tau}{\sigma}\right) d\tau = F(x/\sigma) - F((x - 1)/\sigma).$$

Their ratio is the posterior probability. □

Lemma 2 (Monotone posterior). *Under the log-concavity assumption on f , the posterior probability $\Pr(\tau \leq t \mid x_i = x)$ is nonincreasing in x for every fixed $t \in (0, 1)$. A marginal-indifference solution to equation (1) therefore supports a monotone cutoff best response.*

Proof. For $x' > x$, the likelihood ratio of the two signals is

$$\frac{f((x' - \tau)/\sigma)}{f((x - \tau)/\sigma)}.$$

Log-concavity of f makes this ratio nondecreasing in τ : writing $u = (x - \tau)/\sigma$ and $a = (x' - x)/\sigma > 0$, the ratio $f(u + a)/f(u)$ is nonincreasing in u , while u decreases with τ . The posterior after x' therefore dominates the posterior after x in the monotone-likelihood-

ratio order. Higher signals place more probability on higher thresholds, so $\Pr(\tau \leq t \mid x_i = x)$ is nonincreasing in x . At a solution to equation (1), citizens below the indifferent signal weakly prefer participation and citizens above it weakly prefer abstention. $\square$

A.3 Proof of Proposition 1.

Proof of Proposition 1. Fix a cutoff strategy profile with cutoff $\hat{x}$: citizen i participates if and only if $x_i < \hat{x}$.

Step 1: participation and the induced success threshold. At true threshold τ , the mass of participants is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = \Pr(\tau + \sigma \varepsilon_i < \hat{x}) = \Pr\left(\varepsilon_i < \frac{\hat{x} - \tau}{\sigma}\right) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ . At $\tau = 0$, $N^*(0) > 0$, while at $\tau = 1$, $N^*(1) < 1$. Hence $N^*(\tau) - \tau$ crosses zero exactly once on $(0, 1)$. Denote the crossing by $\hat{\tau}_\sigma$. It satisfies

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \quad (10)$$

and protest succeeds exactly when $\tau \leq \hat{\tau}_\sigma$.

Step 2: reduce the system to one dimension. A citizen with signal x_i gains $\Pr(\tau \leq \hat{\tau}_\sigma \mid x_i)d_p - c_r$ from participating. Lemma 1, evaluated at $t = \hat{\tau}_\sigma$ and $x_i = \hat{x}_\sigma$, gives

$$\Pr(\tau \leq \hat{\tau}_\sigma \mid x_i = \hat{x}_\sigma) = \frac{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - \hat{\tau}_\sigma)/\sigma)}{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - 1)/\sigma)}.$$

Together with marginal indifference, this expression and equation (10) give the exact system in equation (1).

For any candidate threshold $t \in (0, 1)$, define $z(t) = F^{-1}(t)$ and

$$x_\sigma(t) = t + \sigma z(t).$$

This pair satisfies the crossing condition because $F((x_\sigma(t) - t)/\sigma) = F(z(t)) = t$. Substitution into the marginal citizen's posterior gives the continuous function

$$P_\sigma(t) = \frac{F(t/\sigma + z(t)) - t}{F(t/\sigma + z(t)) - F((t-1)/\sigma + z(t))}. \quad (11)$$

An interior solution is a root of $P_\sigma(t) = \rho$, where $\rho = c_r/d_p$.

Step 3: local existence and compactness. Let $t_0 = 1 - \rho$ and fix $\delta \in (0, \min\{t_0, 1 - t_0\})$. On the compact interval $I_\delta = [t_0 - \delta, t_0 + \delta]$, $z(t)$ is bounded. Uniformly for $t \in I_\delta$,

$$F(t/\sigma + z(t)) \longrightarrow 1, \quad F((t-1)/\sigma + z(t)) \longrightarrow 0$$

as $\sigma \rightarrow 0$. Equation (11) therefore implies the uniform convergence

$$\sup_{t \in I_\delta} |P_\sigma(t) - (1 - t)| \longrightarrow 0.$$

At the left endpoint, the limiting value is $1 - (t_0 - \delta) = \rho + \delta$; at the right endpoint it is $1 - (t_0 + \delta) = \rho - \delta$. For all sufficiently small positive σ , continuity then gives

$$P_\sigma(t_0 - \delta) > \rho > P_\sigma(t_0 + \delta).$$

The intermediate value theorem supplies a root $\hat{t}_\sigma \in I_\delta$ for every such σ . Lemma 2 makes the associated cutoff $\hat{x}_\sigma = x_\sigma(\hat{t}_\sigma)$ a best response, so the root is an interior monotone cutoff equilibrium. Its compactness follows from the construction rather than from a separate assumption.

Step 4: convergence of the selected branch. For any selection of roots inside I_δ ,

$$|\hat{t}_\sigma - t_0| = |(1 - \hat{t}_\sigma) - \rho| \leq \sup_{t \in I_\delta} |P_\sigma(t) - (1 - t)| \longrightarrow 0.$$

Because $F^{-1}(\hat{\tau}_\sigma)$ is bounded on I_δ ,

$$\hat{x}_\sigma - \hat{\tau}_\sigma = \sigma F^{-1}(\hat{\tau}_\sigma) \longrightarrow 0.$$

Hence both equilibrium objects converge to $t_0 = 1 - c_r/d_p$.

Step 5: boundary policies. If $c_r > d_p$, the participation gain is strictly negative even when success is certain, so abstention is uniquely optimal. At $c_r = d_p$, the gain is weakly negative and equals zero when success is certain. Universal abstention and universal participation can therefore both be supported as weak equilibria. The selected full-deterrence outcome is the continuous extension $\hat{\tau}(c_r) \downarrow 0$ as $c_r \uparrow d_p$. At $c_r = 0$, participation is weakly dominant, and full participation is the corresponding continuation $\hat{\tau}(c_r) \uparrow 1$ as $c_r \downarrow 0$. $\square$

Remark 1. At positive noise, equation (1) can have additional roots near the boundaries of the common prior. Proposition 1 selects the root in a fixed neighborhood of $1 - c_r/d_p$ whose existence is guaranteed for all sufficiently small noise. Every selection from that neighborhood has the same limit. The state's problem is defined using this limiting branch.¹⁷

A.4 Positive-noise robustness.

The state problem above uses the small-noise limit. To assess whether the crowd-cost result persists when signals remain noisy, let Φ denote the standard normal c.d.f. For a candidate threshold $t \in (0, 1)$, the first equation in (1) implies $x_\sigma(t) = t + \sigma\Phi^{-1}(t)$. Substituting this expression into the marginal citizen's posterior reduces the two-equation system to the one-dimensional residual in equation (11). When the bounded prior generates more than one root, we follow the interior branch specified in Proposition 1.

17. The locally diffuse limiting argument follows the standard global-games logic in Morris and Shin (2003) and Bueno de Mesquita and Shadmehr (2023).

For each selected root $\hat{\tau}_\sigma$, expected protest mass is calculated directly as

$$\mathbb{E}_\tau[N^*(\tau)] = \int_0^1 \Phi\left(\frac{x_\sigma(\hat{\tau}_\sigma) - \tau}{\sigma}\right) d\tau,$$

The state payoff uses this expected mass rather than imposing its limiting equality with the success probability. For $B_s = 3$, $d_p = 3$, $\theta_1 = 1$, and $H(\bar{N}, c_r; \theta_0) = \theta_0 \bar{N}$, the resulting optimal policies are:

σ	c_r^* at $\theta_0 = 1.0$	c_r^* at $\theta_0 = 1.1$
0.05	0.6947	0.7134
0.03	0.6832	0.7013
0.02	0.6778	0.6954
0.01	0.6722	0.6894

At every reported noise level, raising θ_0 from 1.0 to 1.1 raises optimal expected participant harm. Both policy sequences approach their respective analytic limits, 0.6667 and 0.6833, as noise falls. Expected protest mass also approaches the success probability: at $c_r/d_p = 0.2$, the absolute gap falls from 0.0421 at $\sigma = 0.05$ to 0.0084 at $\sigma = 0.01$.

A.5 Proof of Proposition 2.

Proof of Proposition 2. For $c_r \in [0, d_p]$, substitute $\hat{\tau}(c_r) = 1 - c_r/d_p$ into equation (3):

$$u_S(c_r) = \frac{c_r}{d_p} B_s - H\left(1 - \frac{c_r}{d_p}, c_r; \theta_0\right) - K(c_r; \theta_1).$$

Differentiation gives

$$u_c(c_r) = \frac{B_s + H_N}{d_p} - H_c - K_c,$$

$$u_{cc}(c_r) = -\frac{H_{NN}}{d_p^2} + \frac{2H_{Nc}}{d_p} - H_{cc} - K_{cc}.$$

Assumption 5 makes the objective strictly concave on $(0, d_p)$ and gives $u_c(0^+) > 0$. If $u_c(d_p^-) < 0$, continuity of the interior derivative and strict concavity imply a unique root of $u_c(c_r) = 0$ in $(0, d_p)$, which is the unique maximizer. If $u_c(d_p^-) \geq 0$, the objective is increasing up to the full-deterrence corner and is maximized at $c_r = d_p$ on this interval. At every $c_r > d_p$, the clamped threshold and expected protest mass are zero, so $u_s(c_r) = B_s - H(0, c_r; \theta_0) - K(c_r; \theta_1)$ and its derivative is $-H_c(0, c_r; \theta_0) - K_c(c_r; \theta_1) < 0$. Thus the policy characterized in the proposition is globally optimal. This argument uses one-sided derivatives at the kink $c_r = d_p$. $\square$

A.6 Proof of Proposition 3.

Proof of Proposition 3. At an interior solution define¹⁸

$$G(c_r; B_s, \theta_0, \theta_1, d_p) \equiv u_c(c_r) = 0.$$

Because $G_{c_r} = u_{cc} < 0$, the implicit function theorem gives

$$\frac{\partial c_r^*}{\partial B_s} = -\frac{1/d_p}{u_{cc}} > 0, \quad \frac{\partial c_r^*}{\partial \theta_1} = \frac{K_{c\theta_1}}{u_{cc}} < 0,$$

18. For the first-order-condition function G , $G_z \equiv \partial G / \partial z$ fixes every displayed primitive other than z . In particular, $G_{c_r} = u_{cc}$, while G_{d_p} and G_{θ_0} differentiate the first-order condition at a fixed candidate policy c_r . In the limiting state problem, expected mass equals $\hat{\tau} = 1 - c_r/d_p$, so G_{d_p} includes the induced change $\partial \hat{\tau} / \partial d_p = c_r/d_p^2$ at fixed c_r . The implicit-function step then allows the equilibrium policy to adjust.

where $K_{c\theta_1} > 0$. Fixing c_r and differentiating G in d_p gives

$$G_{d_p} = -\frac{B_s + H_N + c_r H_{Nc} - (c_r/d_p) H_{NN}}{d_p^2},$$

and therefore

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + H_N + c_r^* H_{Nc} - (c_r^*/d_p) H_{NN}}{d_p^2 u_{cc}}.$$

This derivative is negative under the condition stated in Proposition 3. For the crowd-cost shifter,

$$G_{\theta_0} = \frac{H_{N\theta_0}}{d_p} - H_{c\theta_0}, \quad \frac{\partial c_r^*}{\partial \theta_0} = -\frac{H_{N\theta_0}/d_p - H_{c\theta_0}}{u_{cc}}.$$

Since $u_{cc} < 0$, this derivative is positive exactly when the numerator is positive. This is the crowd-cost condition. The primitive B_p is absent from G , which gives its zero direct partial effect.

Finally, $\hat{\tau}^* = 1 - c_r^*/d_p$. Its response to B_s , θ_0 , and θ_1 has the opposite sign from the corresponding response of c_r^* . For motivation,

$$\frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{c_r^*}{d_p^2} - \frac{1}{d_p} \frac{\partial c_r^*}{\partial d_p} > 0,$$

because $c_r^* > 0$ and $\partial c_r^*/\partial d_p < 0$. The direct partial effect of B_p on $\hat{\tau}^*$ is also zero. □

A.7 Proof of Proposition 4.

Proof of Proposition 4. With $d_p(c_r) = d_0 + e(c_r)$, apply the quotient rule to $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$:

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_p(c_r) \cdot 1 - c_r e'(c_r)}{d_p(c_r)^2} = -\frac{d_p(c_r) - c_r e'(c_r)}{d_p(c_r)^2}.$$

The denominator is positive, so the derivative has the sign of $-(d_p(c_r) - c_r e'(c_r))$. Dividing the parenthesis by $d_p(c_r) > 0$,

$$d_p(c_r) - c_r e'(c_r) \geq 0 \iff 1 - \frac{c_r e'(c_r)}{d_p(c_r)} = 1 - \varepsilon(c_r) \geq 0,$$

so $\partial \hat{t} / \partial c_r < 0$ exactly when $\varepsilon(c_r) < 1$ and $\partial \hat{t} / \partial c_r > 0$ exactly when $\varepsilon(c_r) > 1$. $\square$

A.8 Proof of Proposition 5.

Proof of Proposition 5. At an interior threshold, equation (9) gives $D = -\hat{t}_c$. Differentiating the state payoff at a fixed α gives the first-order condition

$$u_c(c_r; \alpha) = (B_s + H_N)D - H_c - K_c = 0.$$

At a positive policy, $B_s + H_N > 0$, $H_c \geq 0$, and $K_c > 0$. The equality is therefore impossible when $D \leq 0$. Every positive interior optimum must have $D > 0$, which is equivalent to $\varepsilon(c_r^*; \alpha) < 1$.

Under the deployment-separable technology, the first-order condition becomes

$$G(c_r, \alpha) \equiv (B_s + w)D(c_r, \alpha) - A'(c_r) - K_c(c_r; \theta_1) = 0.$$

Holding c_r fixed and differentiating the marginal deterrent return in α gives

$$D_\alpha(c_r, \alpha) = -\frac{e(c_r)[d_p(c_r; \alpha) - c_r \alpha e'(c_r)] + c_r e'(c_r) d_0}{d_p(c_r; \alpha)^3} < 0$$

whenever the emotional response is nonzero and $D > 0$. Hence $G_\alpha = (B_s + w)D_\alpha < 0$. Because $G_{c_r} = u_{cc} < 0$ at the strictly concave interior optimum, the implicit function theorem yields

$$\frac{\partial c_r^*}{\partial \alpha} = -\frac{G_\alpha}{G_{c_r}} < 0.$$



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